

SQL & DATABASES · DAY 2 OF 10

Building tables

CREATE TABLE, rules (constraints) and foreign keys

Tuesday 6 October 2026

Today

- After today you can:
 - Create a table with CREATE TABLE
 - Choose good data types for columns
 - Add rules: PRIMARY KEY, NOT NULL, UNIQUE, CHECK, DEFAULT
 - Connect tables with foreign keys
 - Change and delete tables with ALTER TABLE and DROP TABLE

Today's schedule

Time	What we do
09:00-09:10	Warm-up
09:10-09:50	Theory (slides)
09:50-10:00	Break
10:00-10:55	Code-along, part 1
10:55-11:05	Break
11:05-12:00	Code-along, part 2
12:00-13:00	Lunch
13:00-14:30	Lab, part 1
14:30-14:45	Break
14:45-16:15	Lab, part 2
16:15-17:00	Walkthrough and wrap-up

Two kinds of SQL

	DDL	DML
Stands for	Data Definition Language	Data Manipulation Language
Simple meaning	Build the cupboard	Put things in and take them out
Commands	CREATE, ALTER, DROP	SELECT, INSERT, UPDATE, DELETE
When	Today	Tomorrow

CREATE TABLE

- Table name first
- Then each column: name + data type
- Commas between columns, but not after the last one

```
CREATE TABLE pets (  
    pet_id  INTEGER PRIMARY KEY,  
    name    TEXT,  
    species TEXT,  
    age     INTEGER  
);
```

Choosing data types

Information	Data type	Why
Age	INTEGER	Whole number
Price	REAL	Can have decimals
Name, email	TEXT	Words
Date	TEXT as 'YYYY-MM-DD'	Sorts correctly
Phone number	TEXT	Starts with 0, never do math with it

Constraints = rules for your table

Rule	Means	Like...
PRIMARY KEY	Unique ID for each row	Personal number
NOT NULL	Must have a value	Name field marked with *
UNIQUE	No two rows can be the same	Two people can't share an email
CHECK	Value must pass a test	Age can't be negative
DEFAULT	Value used if you give none	Status starts as 'new'

Rules in code

```
CREATE TABLE members (  
  member_id INTEGER PRIMARY KEY,  
  name      TEXT NOT NULL,  
  email     TEXT UNIQUE,  
  age       INTEGER CHECK (age >= 0),  
  level     TEXT DEFAULT 'beginner'  
);
```


Tables need to talk to each other

- An order belongs to a customer
- We don't copy the customer's name into every order
- Instead we write the customer's ID: `customer_id = 1`
- That's like writing 'see page 12' instead of copying page 12

order_id	customer_id	order_date
1	1	2026-01-05
3	1	2026-01-20

Foreign key: a pointer to another table

- The database checks that customer 1 really exists

```
CREATE TABLE orders (  
  order_id      INTEGER PRIMARY KEY,  
  customer_id   INTEGER NOT NULL,  
  order_date    TEXT NOT NULL,  
  FOREIGN KEY (customer_id)  
    REFERENCES customers(customer_id)  
);
```

Our webshop after today

Table	What it holds	Points to
customers	Who buys	-
products	What we sell	-
orders	One shopping trip	customers
order_items	Each product in an order	orders and products

A key made of two columns

- In order_items, order_id alone is not unique (an order has many items)
- product_id alone is not unique (a product is sold many times)
- But the pair is unique: each product once per order

```
PRIMARY KEY (order_id, product_id)
```

Changing and removing tables

- ALTER TABLE: change an existing table
- DROP TABLE: delete the whole table, with all its data
- There is no undo button in real life. Be careful with DROP!

```
ALTER TABLE pets ADD COLUMN owner TEXT;
```

```
DROP TABLE pets;
```

Remember

- DDL builds tables: CREATE, ALTER, DROP
- Every column needs a data type
- Constraints are rules: PRIMARY KEY, NOT NULL, UNIQUE, CHECK, DEFAULT
- A foreign key points to a row in another table
- Press Ctrl+S (Write Changes) to save!

Your turn (1/2)

1. Create a table books with: book_id (primary key), title (must have a value), author, year (whole number).
2. Add a rule to books so year must be greater than 1400. (Hint: DROP and CREATE again.)
3. Add a column isbn to books. It should be TEXT.
4. Delete the books table.
5. Create a table reviews: review_id, product_id (points to products), rating (1 to 5), comment.

Your turn (2/2)

6. Test reviews: try to add a review with rating 6. What happens?

7. Test reviews: try to add a review for product 50. What happens?

8. On paper: draw the 4 webshop tables as boxes and draw arrows for each foreign key.

Coming up tomorrow

Our tables are empty. Tomorrow we fill them with data, change it and delete it!